

Introduction to Tilings

Let $\mathcal{T} = \{T_\alpha\}_{\alpha \in A}$ be a collection of subsets of the Euclidean plane $\mathbb{R}^2$. We say that $\mathcal{T}$ is a **tiling (or tessellation)** if the following three conditions hold:

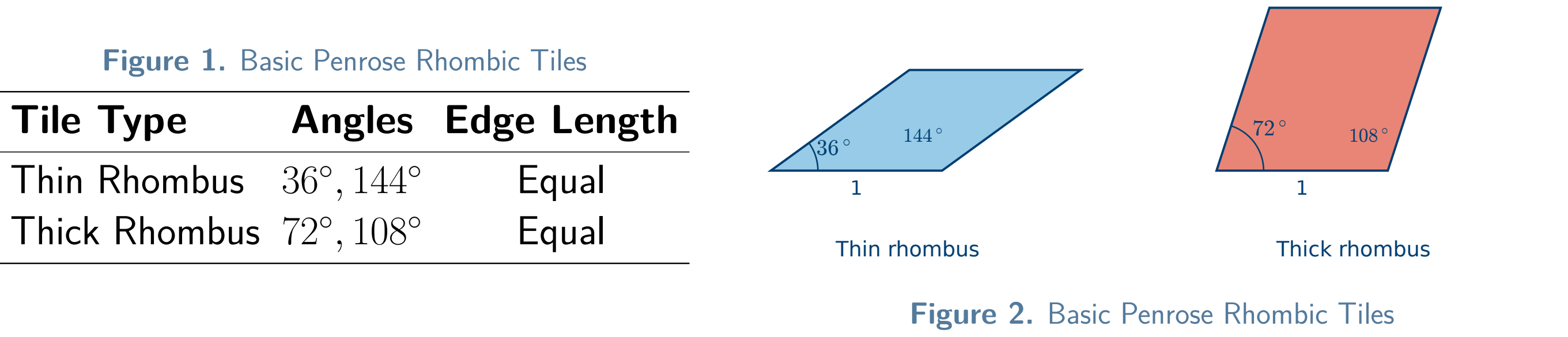
1. **Covering (no gaps):**
- $$\bigcup_{\alpha \in A} T_\alpha = \mathbb{R}^2.$$
2. **Non-overlap:** for every $\alpha \neq \beta$,
- $$\text{int}(T_\alpha) \cap \text{int}(T_\beta) = \emptyset,$$

where $\text{int}(\cdot)$ denotes the interior.
3. **Topological disks:** each tile T_α is *homeomorphic* to the closed unit disk
- $$D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$

Fundamentals of Penrose Tiling

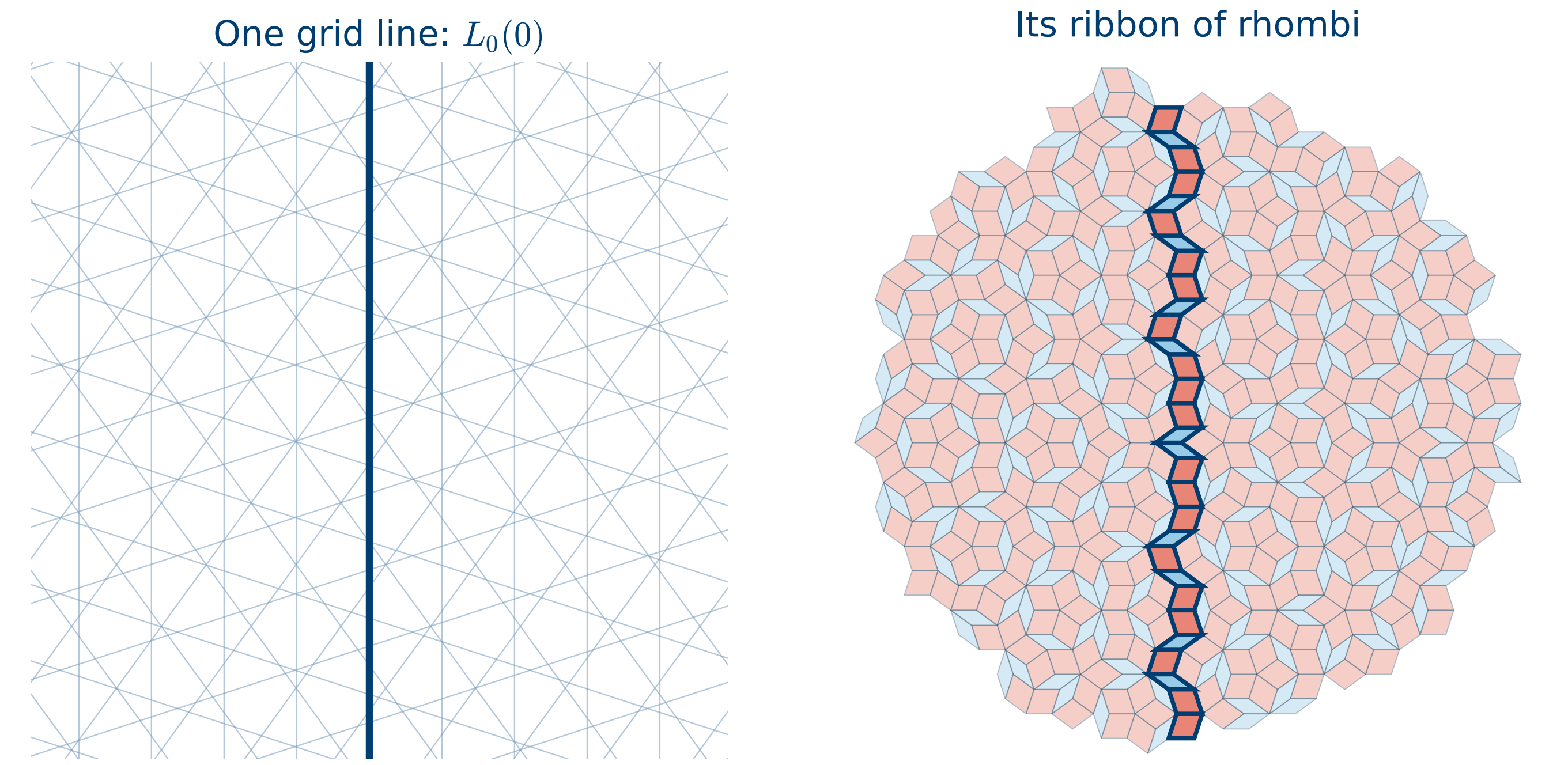
Penrose tilings are **non-periodic tilings** discovered by mathematician and physicist Roger Penrose in the 1970s. Unlike regular periodic tilings (like squares or hexagons), Penrose tilings:

- **No Translational Symmetry** – no nonzero translation preserves the entire tiling; finite patches do recur.
- **Can be constructed using different rules** – for example, matching rules and projection methods (e.g., de Bruijn’s pentagrid approach)



Pentagrid Patterns

Each rhomb has **two pairs of parallel edges**, you can chain rhombs edge-to-edge along any one pair and **create a long “ribbon”**. There are only **five distinct edge directions** in the Penrose rhomb set, so **ribbons within the same family do not intersect with each other** .



de Bruijn found that the ribbons can be reconstructed into sets of parallel lines, which is called **pentagrid**. While the **intersections of 2 lines** in the pentagrid pattern can be represented a **particular rhomb**.

Constructing Penrose Tilings with Pentagrid

1. **Five grids** For $j = 0, \dots, 4$, set the unit direction vectors

$$v_j = (\cos 2\pi j/5, \sin 2\pi j/5), \quad \gamma_0 + \dots + \gamma_4 = 0.$$

The j -th line family is

$$\mathcal{L}_j = \{x \in \mathbb{R}^2 \mid x \cdot v_j + \gamma_j \in \mathbb{Z}\}.$$

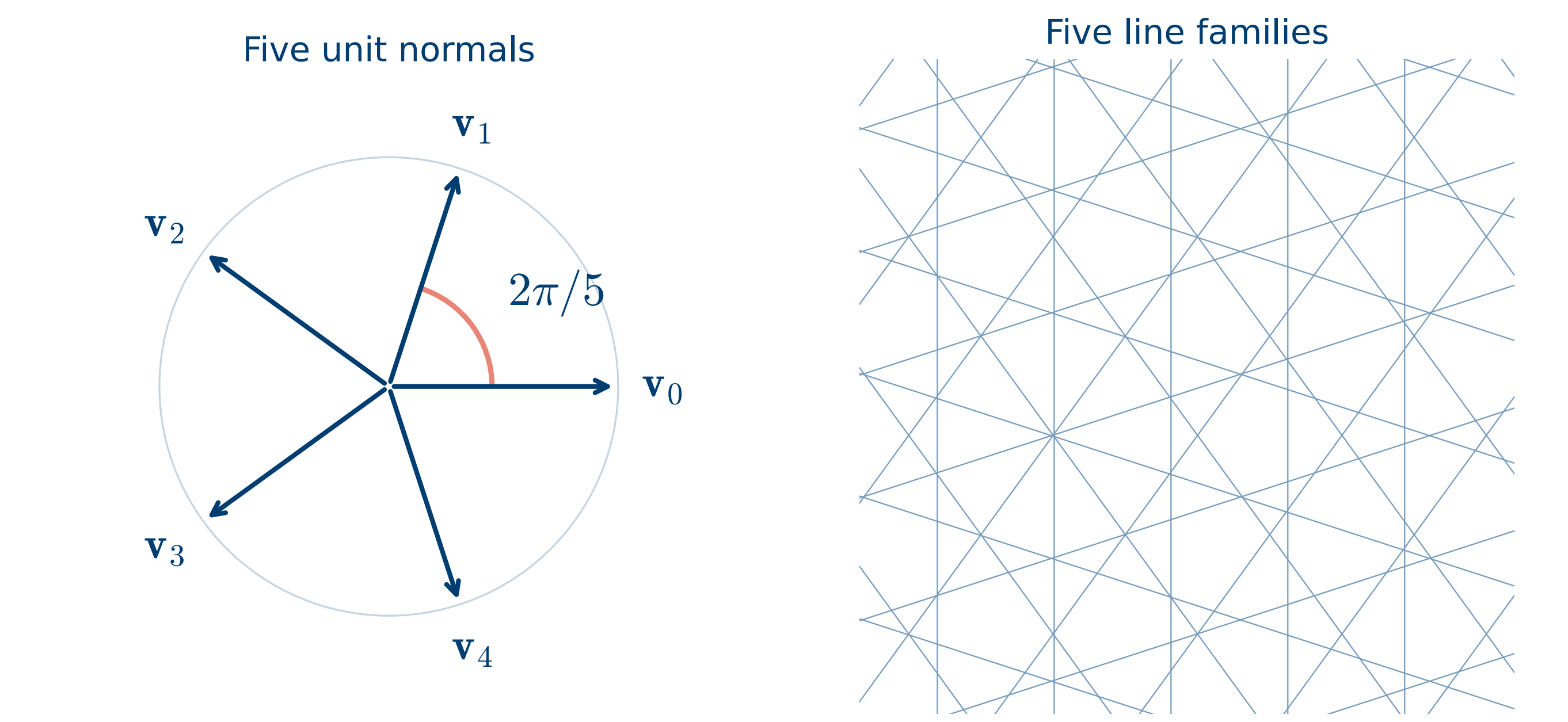


Figure 4. de Bruijn’s Pentagrid and its Corresponding Vectors. Adjacent vectors are separated by $2\pi/5$.

The pentagrid is called **regular** if no point of $\mathbb{R}^2$ belongs to **more than two of the five grids**, and otherwise singular.

2. **Integer coordinates of a cell** For any $x \in \mathbb{R}^2$, define the integer-valued functions

$$K_j(x) = \lceil x \cdot v_j + \gamma_j \rceil, \quad K(x) = (K_0, \dots, K_4) \in \mathbb{Z}^5.$$

where: γ_j are the grid shift parameters

$K(x) = (K_0(x), ..., K_4(x))$ is a five-tuple, which are representing the **“pentagrid coordinates of a vertex in the tiling”**.

The map $x \mapsto K(x)$ is **constant** on each open cell and they change only when **crossing a grid line**.

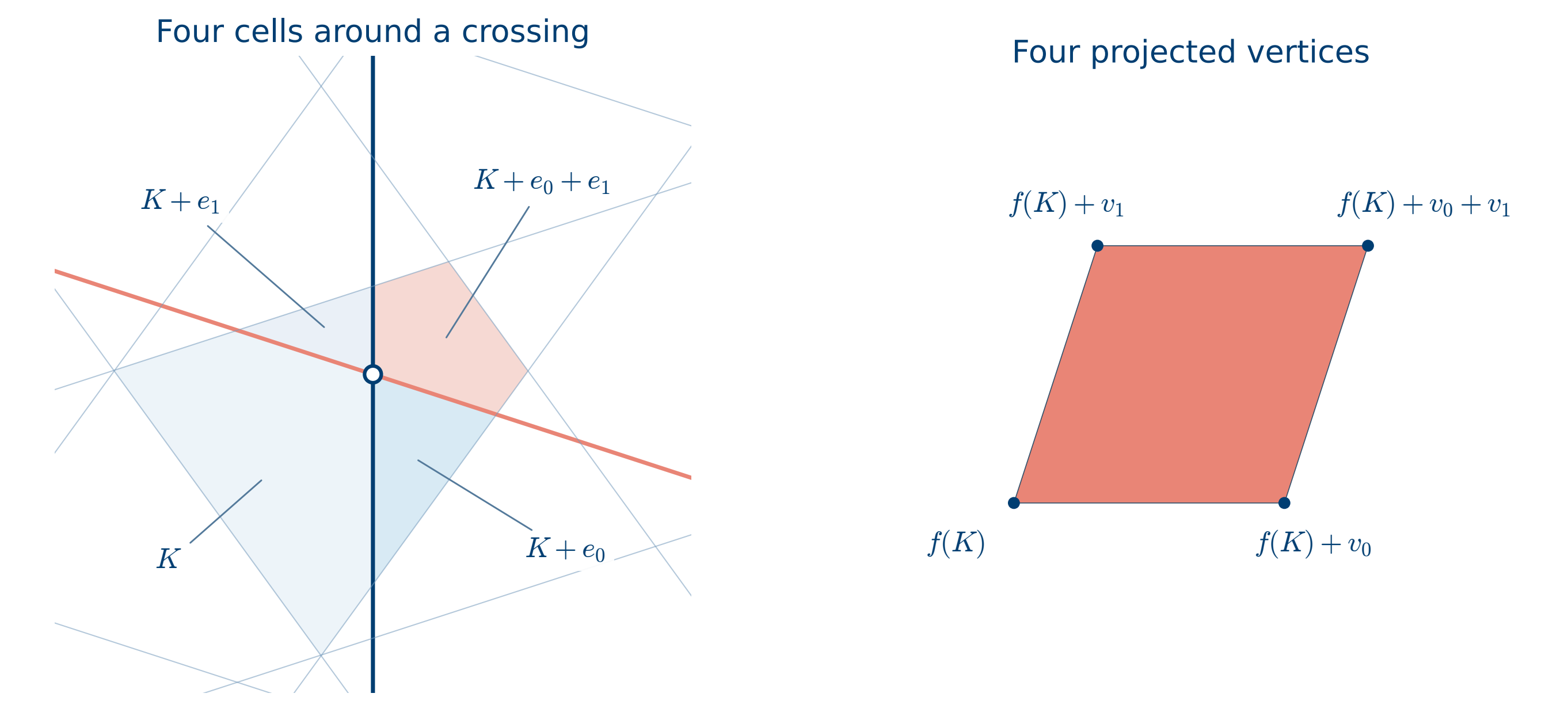


Figure 5. Coordinates in Pentagrid. Four neighbouring cell labels project to the vertices of a rhombus.

3. **Linear projection to the plane**

$$f(k_0, \dots, k_4) = \sum_{j=0}^4 k_j v_j$$

The $K(x)$ act as “coordinates” in a 5D space, and $f(K(x))$ projects them to 2D. The resulting point $f(K(x))$ is a vertex in the Penrose tiling.

(Continue) Constructing Penrose Tilings with Pentagrid

4. **Each two-line crossing gives one rhomb.** Choose $0 \leq r < s \leq 4$ and integers k_r, k_s . Solving

$$v_r \cdot x + \gamma_r = k_r, \quad v_s \cdot x + \gamma_s = k_s$$

gives an **intersection** x_0 .

For example, given the **five-tuple** $K(x_0) = (K_0, \dots, K_4)$ and the pair $(r, s) = (0, 1)$, the **four vertices** correspond to

$$K(x_0) + (\varepsilon_1, \varepsilon_2, 0, 0, 0), \quad \varepsilon_1, \varepsilon_2 \in \{0, 1\}.$$

Thus, we apply **the linear map** $f(k) = \sum_{j=0}^4 k_j v_j$ to each 5–tuple, which gives

$$f_{\varepsilon_1, \varepsilon_2} = f(K(x_0)) + \varepsilon_1 v_r + \varepsilon_2 v_s, \quad (r, s) = (0, 1).$$

Hence, we have $f_{1,0} - f_{0,0} = v_r$, $f_{0,1} - f_{0,0} = v_s$, so the four vertices (images of the 5-tuple coordinates)

$$\{f_{0,0}, f_{1,0}, f_{1,1}, f_{0,1}\}$$

can **form a parallelogram with edge vectors** $\{v_r, v_s\}$.

Therefore, we can determine whether it is a **thin rhomb** or a **thick rhomb** by computing the angle between v_r and v_s

- If angle is 36° or 144°, then the vertices form a **thin rhomb**
- If angle is 72° or 108°, then they form a **thick rhomb**

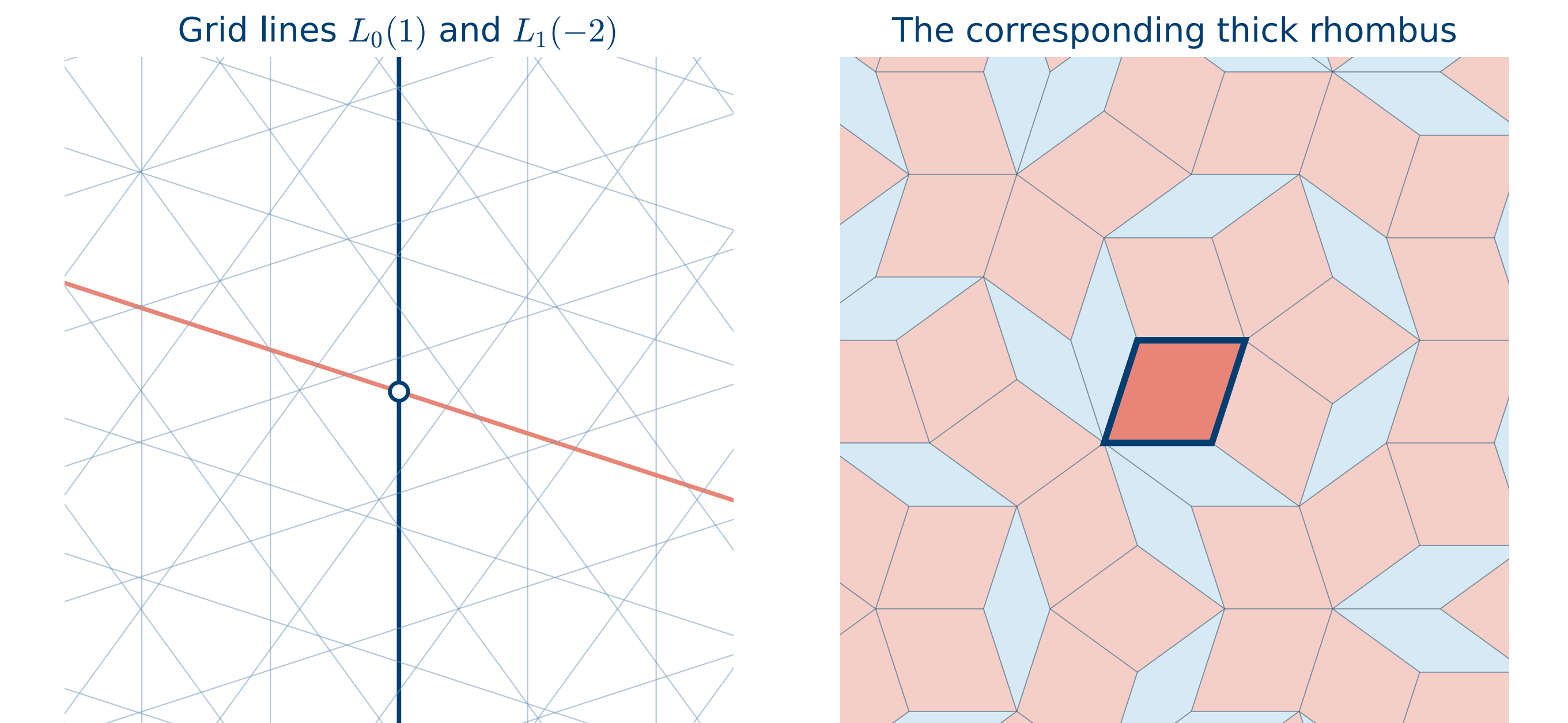


Figure 6. Intersection of two lines with its corresponding rhomb.

cell $K(x)$

$\longleftrightarrow$

tiling vertex $f(K)$

intersection of two grids

$\longleftrightarrow$

single rhomb

Conclusion

Penrose Tilings can be constructed with different methods and Pentagrid approach is discovered by de Bruijn. The illustrations on the poster are not the only one examples. In fact, there are actually infinitely **(indeed uncountably)** many distinct Penrose tilings, because **varying the grid shifts while preserving their zero sum and regularity** changes the arrangement of tiles but still producing a "legal" Penrose tiling.

References

[1] A. Treibergs, Penrose Tiling (lecture slides). University of Utah. PDF, www.math.utah.edu/~treiberg/PenroseSlides.pdf.
[2] E. Mowry & B. Shukla, Pentagrids and Penrose Tilings (Hudson River Undergraduate Math Conf.). Williams College, 2016. PDF.
[3] L. Effinger-Dean, The Empire Problem in Penrose Tilings. Williams College, 2006. Advisor: D. Bailey. PDF.
[4] N. G. de Bruijn, "Algebraic theory of Penrose's non-periodic tilings of the plane I, II", Indagationes Math. 43 (1981) 39–52 and 53–66.
[5] A. Bhatia, "Pattern Collider" (interactive web app). <https://aatishb.com/patterncollider/> (accessed 11 Jun 2025).